

Exercise 2

N.B: Unit Testing has been conducted and validated on the project called Triangle testing.

Task 1 – Example Applications for Software Quality Attributes:

Find two example applications for each of the Software Quality attributes listed below. One of the examples should not be a software application, but a physical object instead (fridge, car, pen, ...).

Functional: an engine inside a car, a keyboard in a laptop

Suitability: The Car's engine provides enough power to drive the car at highway speeds.

The Laptop's word processor allows users to write and edit documents efficiently.

Reliability: brakes always work when you press them, the keyboard buttons should do it's required job whenever you hit the button.

Usability: the brakes should be placed in a place where the user expect it to be, the keyboard buttons should be also placed where the user expects them to be.

Security: if the engine heat turns up it should warn the user. The laptop shall enforce user to insert their correct password when they turn it on

Compatibility: the engine should be able to work with the other components around it. The keyboard should fit into the desing of the laptop.

Portability: the car's light pulps should support change. The laptop's OS shall support updates

Maintainability: the car should be designed in a way that let's the mechanic easily understand the problems. The computer should have architecture that supports upgrade in its components

Performance: The car can accelerate from 0–60 mph in under 6 seconds. The software can open a file in under 2 seconds.

Efficiency: the car should be able to as fast as it can using the least amount of gas possible. The computer should run as many programs as possible fast with low usage of ram and cpu.

Explaining Performance and Suitability:

Performance: focuses on how fast the system responds and the capacity it can deal with. Unlike efficiency which cares more about how much resource does it take to achieve this performance.

Suitability: Does this system provide the right functions so the user can do what they needs, it does not care about the resources or the performance it cares about the function.

Task 2 – Importance of Software Quality Attributes:

Describe two reasons as to why it is important to talk about Software Quality Attributes and why they should be prioritized: Discussing quality attributes ensures that both technical and non-technical stakeholders know what good quality means in the system it guides architecture and implementation. When we prioritize we ensure to direct our limited resources towards what matters the most for our system. team can then make some trade off based on what is more important.

How could you use ISO 25010 in a practical application: ISO 25010 can be used as a framework and reference to compare a system against standardized software quality attributes, helping teams understand what “good quality” means.

Have a software’s good/poor quality ever been exciting/frustrating for you? Name one positive and one negative example: The consistent change in the UI of Instagram is unnecessary it provides no value and derives away from what similar apps are.